

CC-112 Programming Fundamentals Lab 06

Question 1: Find the Second Smallest Element in an Array

Write a C program that takes an array of integers as input and finds the second smallest element in the array.

Sample Input:

Enter the number of elements: 5
Enter elements: 3 1 5 2 4

Sample Output:

The second smallest element is: 2

Question 2: Bubble Sort and Time Calculation

Write a C program that implements the Bubble Sort algorithm to sort an array of integers in ascending order. Also, use the `clock()` function to measure the execution time of the sorting process.

Sample Input:

Enter the number of elements: 5
Enter elements: 5 2 9 1 3

Sample Output:

Sorted array: 1 2 3 5 9
Execution time: 0.000002 seconds (One Example)

Question 3: Most Frequent Character in a String

Write a C program that takes a string as input and finds the most frequently occurring character in the string along with its frequency count.

Sample Input:

Enter input : programming

Sample Output:

Most frequent character: g m
Frequency: 2